

Hafiz Abdul Rehman

AI & Machine Learning Engineer

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Professional Summary

AI & Machine Learning Engineer and Data Scientist with 2+ years of experience across the complete data lifecycle, from data collection and preparation to model training, evaluation, and deployment. Specialized in Machine Learning, Deep Learning, Computer Vision (YOLO, OCR, segmentation), and RAG-based LLM applications such as chatbots and document intelligence. Strong research background in medical imaging and remote sensing, with a focus on building interpretable models that solve real business problems.

Education

GIFT University

Bachelor of Science in Data Science | CGPA: 3.65/4.0

2022 – 2026

Gujranwala, Pakistan

- Focus: Machine Learning, Deep Learning, Computer Vision, NLP, Statistical Analysis

Experience

AI & Machine Learning Engineer (Intern — Onsite)

Apr 2025 – Dec 2025

Neel Tech (Software House)

Pakistan

- Developed RAG-based chatbots and LLM applications for enterprise document intelligence and automation
- Designed and trained ML and Deep Learning models with end-to-end pipelines, improving scalability, accuracy, and robustness
- Collaborated with cross-functional engineering teams to optimize data pipelines and deliver data-driven solutions

AI/ML Consultant & Developer (Freelance)

2025 – Present

Self-Employed (Upwork)

Remote

- Delivering Machine Learning and Computer Vision solutions for international clients: prediction models, YOLO-based detection, OCR data extraction, and image segmentation
- Managing complete project workflow from data annotation and model training to evaluation and documented delivery

Deep Learning & AI Researcher

2025 – Present

GIFT University

Gujranwala, Pakistan

- Developed attention-based U-Net for 3D brain tumor segmentation on BraTS 2020 MRI dataset, achieving 99.64% validation accuracy and 0.986 IoU
- Built hybrid CNN-TDA (Topological Data Analysis) models for medical image classification with improved interpretability

Projects

Customer Churn Prediction & Business Forecasting | Scikit-learn, XGBoost, Pandas

- Built ML pipelines for churn prediction, sales forecasting, and anomaly detection to support business decision-making

Human Activity Recognition System | CNN-LSTM, Deep Learning

- Developed deep learning system recognizing human activities from video sequences using hybrid CNN-LSTM architecture

Application Tracking System using RAG | LangChain, LLMs, Vector DB

- Built intelligent ATS chatbot for automated resume screening and candidate ranking using retrieval-augmented generation

Fire & Smoke Detection for Drone Surveillance | YOLO, PyTorch

- Trained YOLO model on 22,000+ images for real-time aerial fire/smoke detection and alerting

PPE Safety Compliance System | YOLO, OpenCV

- Built helmet and safety-vest detection from live CCTV feeds, achieving 91% precision for industrial safety monitoring

Invoice Data Extraction (OCR Pipeline) | EasyOCR, PaddleOCR, Python

- Developed OCR pipeline converting scanned invoices into structured Excel output for business automation

Factory Quality Control & Defect Detection | CNN, Transfer Learning

- Built image classification and inspection system detecting product defects on a moving conveyor belt

Research Work

- **Survey Paper (Ongoing):** A Comprehensive Survey of Referring Remote Sensing Image Segmentation — From Supervised Cross-Modal Learning to Multimodal Foundation Models
- **Research Paper (Ongoing):** Enhancing Referring Remote Sensing Image Segmentation through Optimal Transport Alignment and Segment Anything Model (SAM)
- **Research Paper:** COVID-19 Detection from CT Scan Images using Topological Data Analysis and CNN Ensemble

Achievements & Honors

- **National Skills Competency Test (2026):** Achieved 89 percentile rank among 34,000+ participants across Pakistan
- **Top 100 Globally — USAII Global AI Hackathon:** Built **CogniFlow**, an AI “Second Brain” decision-support system that estimates student cognitive load, retrieves relevant notes, and recommends prioritized action plans
- **lablab.ai — Bright Data AI Agents Hackathon:** Built **VendorSentinel**, an AI-powered vendor risk intelligence platform with real-time web data and an AI risk scoring engine
- **lablab.ai — Transforming Enterprise Through AI:** Built **NeuralNexus**, a multi-agent financial intelligence system that monitors data, detects contradictions, and surfaces insights proactively
- **Harvard Health Systems Innovation Lab (HSIL), 7th Edition (Onsite):** Developed an AI mental health monitoring platform with digital phenotyping, PHQ-9/GAD-7 mood tracking, and CBT-based interventions
- **ICSC Competition:** Solved applied real-world problem-solving challenges

Technical Skills

Programming: Python, SQL, R, C++, Java

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Feature Engineering, Model Evaluation

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, LSTM, Transformers, Transfer Learning, Fine-Tuning

Computer Vision: OpenCV, YOLOv8/v11, U-Net, SAM, DeepSORT, ByteTrack, EasyOCR, PaddleOCR, Tesseract

LLMs & RAG: LangChain, RAG Systems, Chatbots, Hugging Face, Prompt Engineering, Vector Databases

Data Science: Pandas, NumPy, Matplotlib, Statistical Analysis, Data Visualization, Power BI

Tools & Databases: Git/GitHub, Docker, FastAPI, Streamlit, MySQL, PostgreSQL, MongoDB

Certifications

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| • Agentic AI and AI Agents: A Primer for Leaders | • Introduction to Computer Vision and Image Processing |
| • Programming with Generative AI (IIT) | • Deep Learning with PyTorch |
| • Applied Data Science with Python | • AI Capstone Project with Deep Learning |
| • Machine Learning with Python | |